

Quantum Random Walks for Market Microstructure

Phase 6 final presentation BTCUSDT, June 12, 2026

Research Question

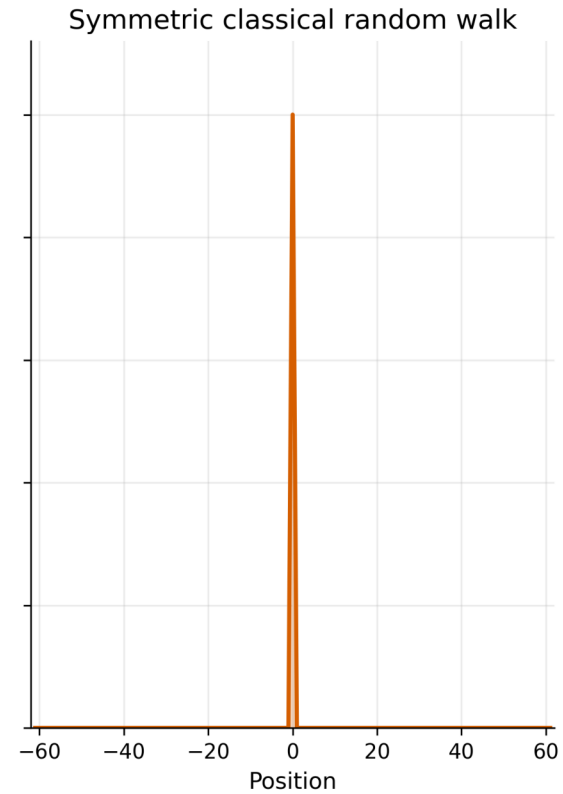
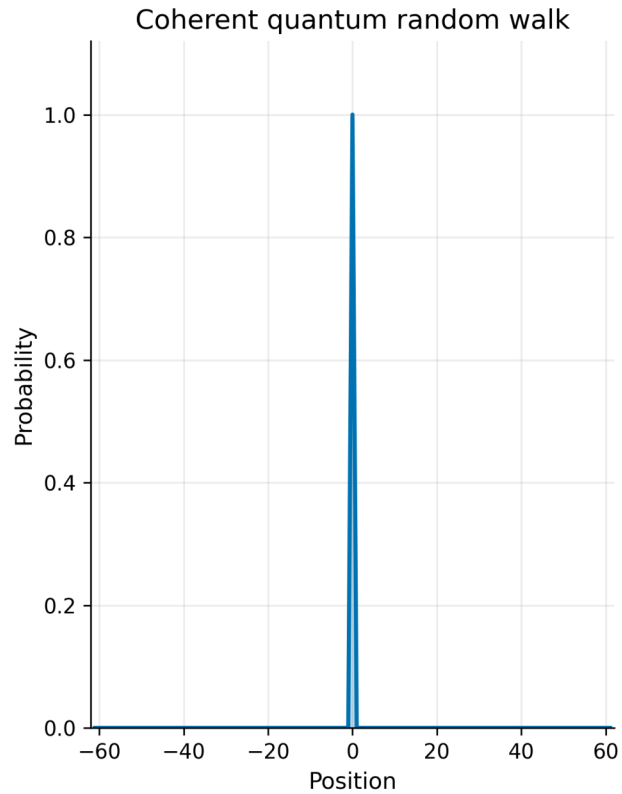
Can a causally calibrated QRW reproduce short-horizon market properties better than classical baselines?

Why QRW?

Interference, ballistic spreading, and a tunable coherent-to-diffusive transition.

QRW vs CRW

Probability evolution at $t = 0$



The coherent walk spreads with interference peaks; the classical walk follows a diffusive binomial envelope.

Active Data

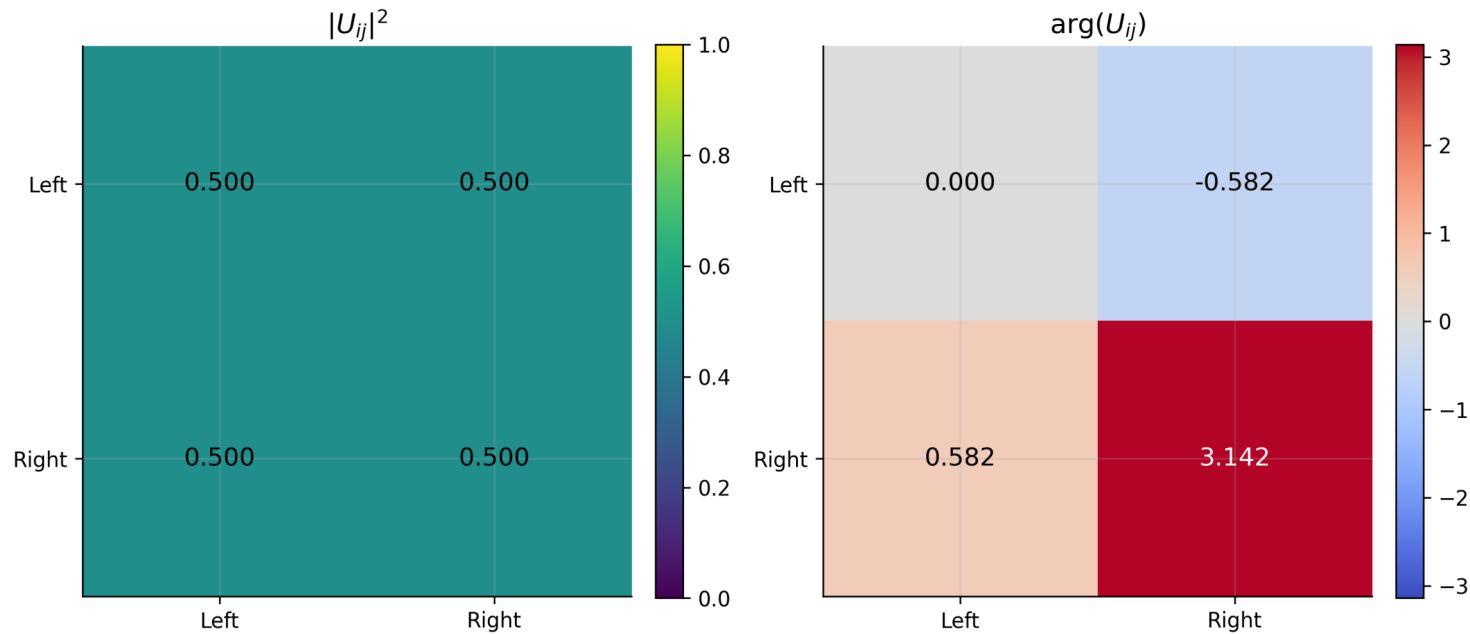
1,869,214 training rows 1,246,144 later holdout rows Historical
June 1-7 derived artifacts remain invalidated.

Causal Pipeline

Trailing cleaner -> chronological split -> disjoint validation
-> fixed seed -> matched statistical samples

Market Mapping

OBI-adaptive coin at OBI=0.75, alpha=0.82
Directional information is phase-encoded; magnitudes remain unitary.



OBI and direction enter a unitary phase-adaptive coin. Trade intensity controls basis dephasing.

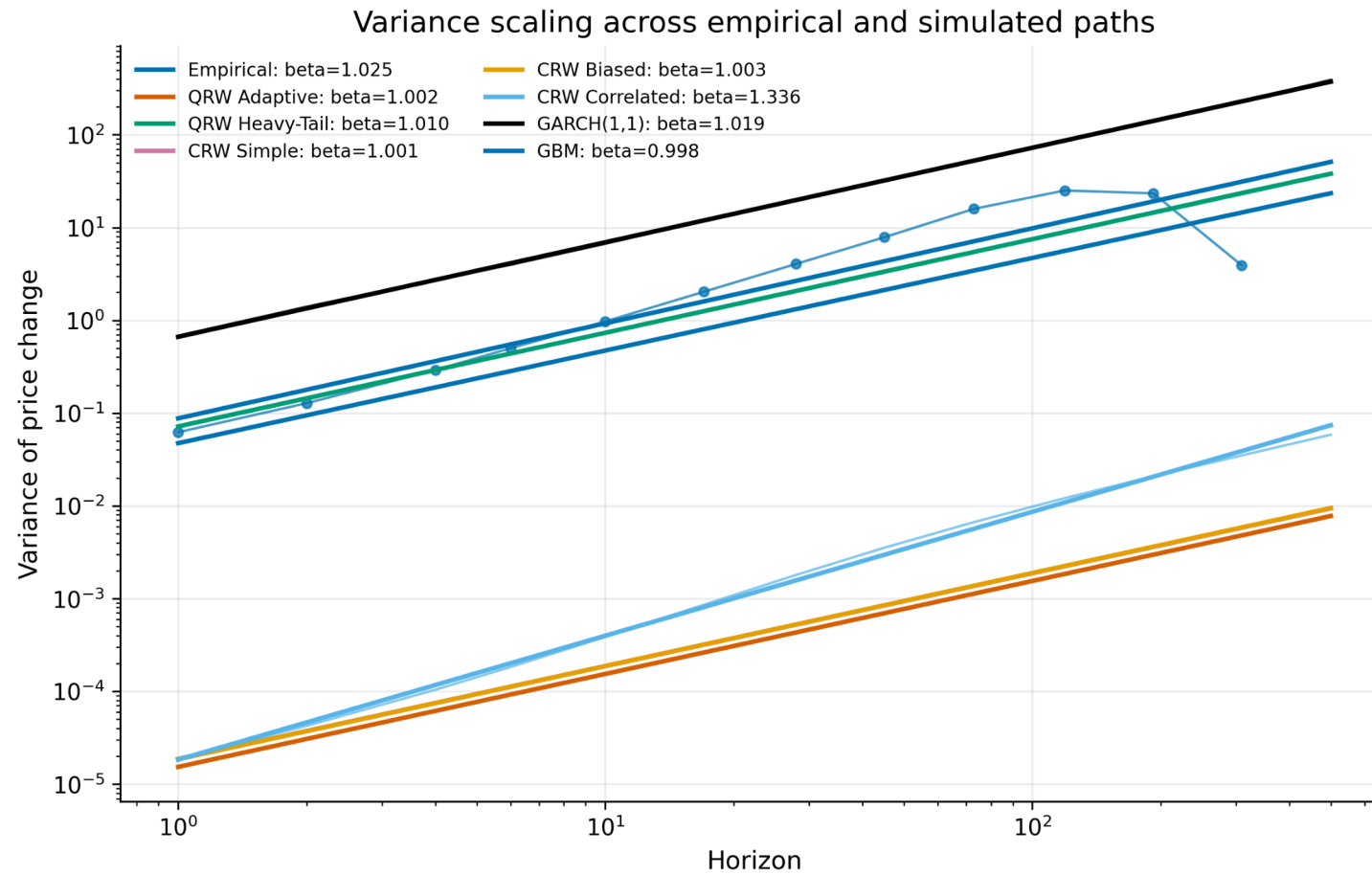
Benchmark Models

QRW Adaptive CRW Simple / Biased / Correlated GARCH(1,1) GBM

Evaluation

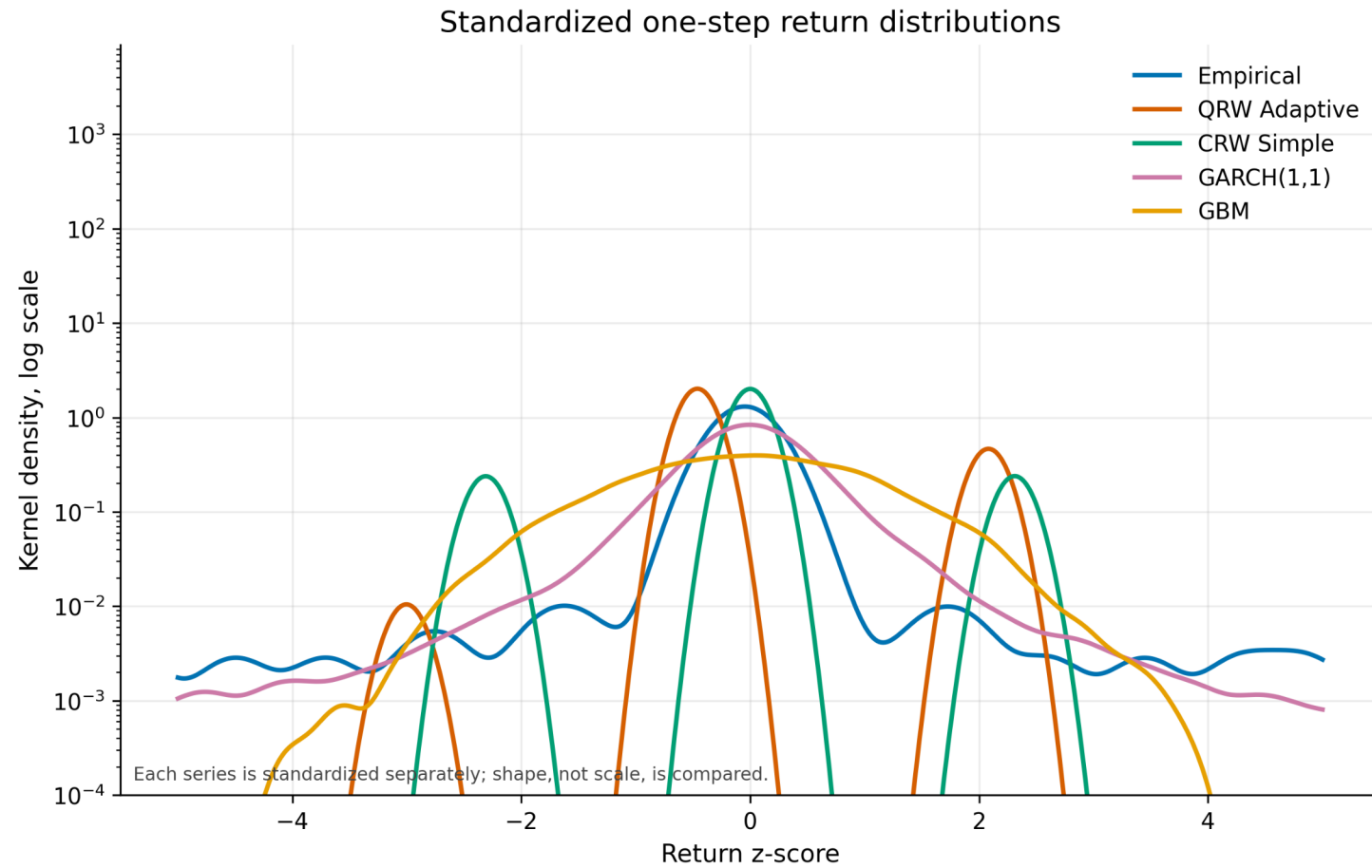
Distribution tests Variance scaling Autocorrelation Tail risk
Eight-metric scorecard

Variance Scaling



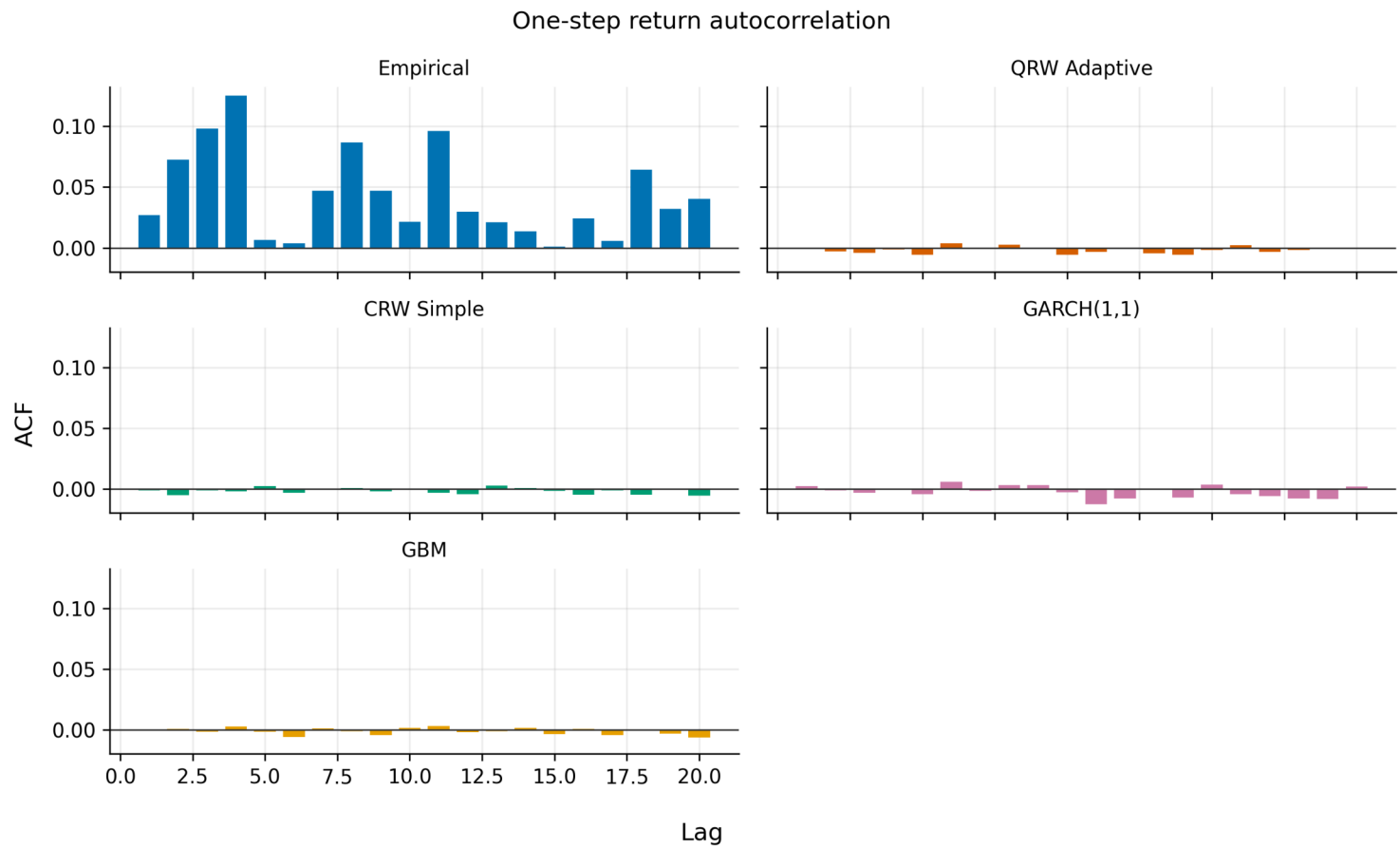
QRW $\beta = 1.0026$ [0.9841, 1.0183]

Distribution Shape



QRW adjusted KS p-value = $9.553e-04$

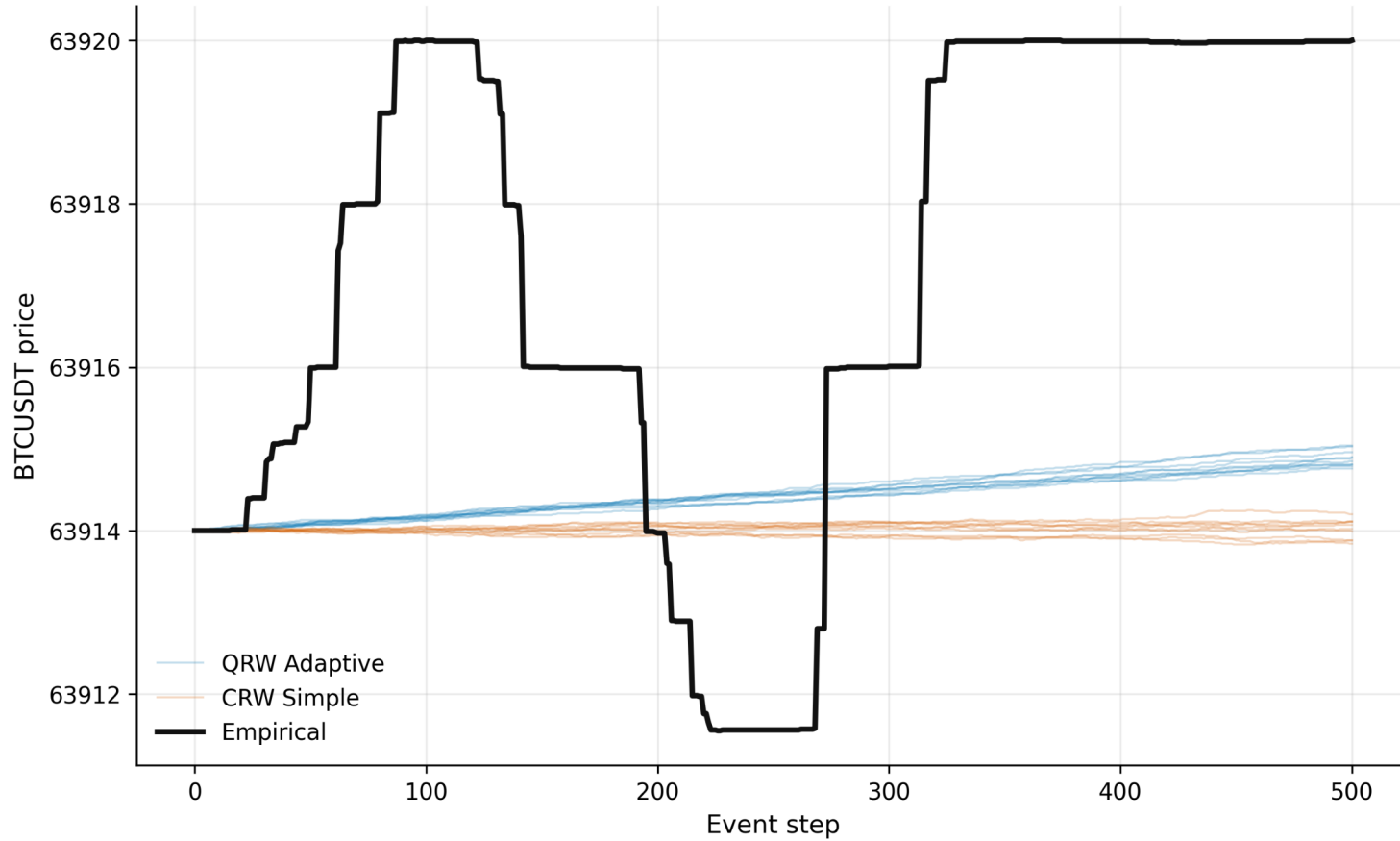
Autocorrelation



QRW ACF MSE = 0.003275

Sample Paths

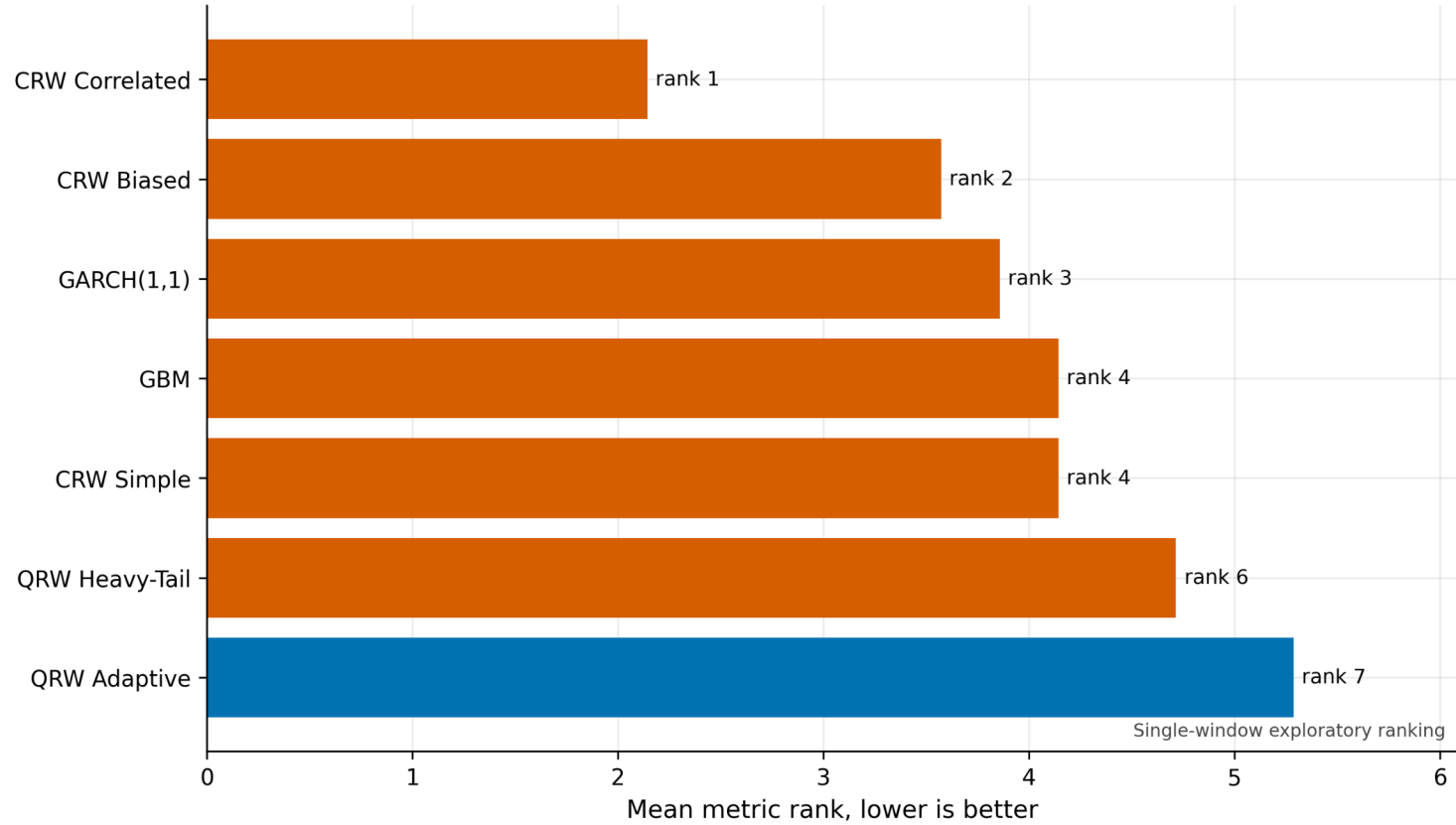
Empirical path and simulated QRW/CRW paths



Path overlays expose scale and directional differences hidden by aggregate ranks.

Scorecard

Phase 5 benchmark scorecard



CRW Correlated ranks first. QRW Adaptive ranks 7.

What We Can Claim

The engineering and reporting pipeline passes. Current evidence does not establish QRW predictive or distributional superiority.

Next Confirmatory Study

Freeze the protocol, collect at least 20 fresh UTC days with synchronized LOB data, then execute one untouched evaluation.